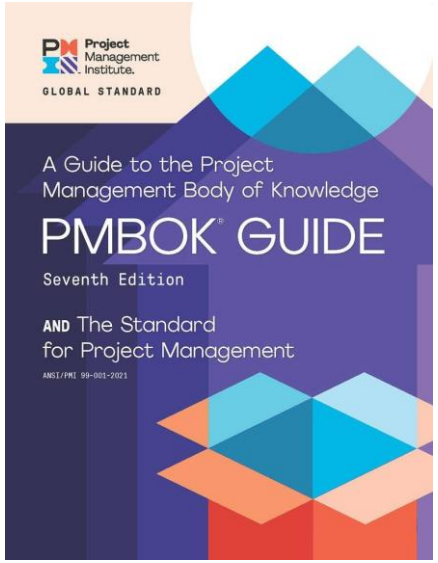
IT3120- Lecture 10

Project Management – Scope & Cost Management

Activities, Tools, Techniques, Inputs and Outputs related to Scope Management and Cost Management



Lecture Objectives

- Explain the importance of Project Scope Management and Project Cost Management.
- Identify activities/processes involved in scope and cost management.
- Describe Inputs, Tools & Techniques, and Outputs (ITTOs).
- Apply scope and cost management concepts to IT industry projects.
- Understand practical project management challenges in software projects.

What is Scope Management?

- Project Scope Management refers to the processes and activities required to ensure that a project includes **all the work needed, and only the work needed**, to successfully complete the project.



- Scope management focuses on defining and controlling what is included and excluded from the project. It helps project teams clearly understand project boundaries, deliverables, requirements, and stakeholder expectations.

Main Objectives of Scope Management

- Define project and product requirements clearly
- Establish project boundaries
- Prevent scope creep (uncontrolled changes)
- Ensure deliverables meet stakeholder expectations
- Improve project planning, estimation, and control



Project Scope Vs Product Scope

- Projects must clearly distinguish between **Project Scope** and **Product Scope** to ensure successful delivery and effective control of project work.



Project Scope

- **Project Scope** refers to the **work that must be performed** to deliver a product, service, or result with the specified features and functions.
- It focuses on:
  - Tasks and activities
  - Deliverables
  - Resources and effort required
  - Project boundaries



Product Scope

- **Product Scope** refers to the **features, characteristics, and functions** of the product, service, or result being created.
- It focuses on:
  - Functional requirements
  - Technical specifications
  - Quality characteristics
  - Product capabilities

Difference Between Project Scope and Product Scope

Aspect	Project Scope	Product Scope
Focus	Work required	Features/functions required
Concern	Project activities and deliverables	Product characteristics
Question Answered	“How will the work be completed?”	“What should the product provide?”
Managed By	Project Manager	Business Analysts / Product Owners / Technical Team
Measured Against	Project plan	Product requirements

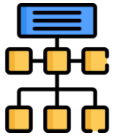
Key Processes in Scope Management

- Following are the major scope management activities:


Plan Scope Management


Collect Requirements


Define Scope


Create Work Breakdown Structure (WBS)


Validate Scope


Control Scope

Key Processes in Scope Management



Plan Scope Management

Plan Scope Management is the process of creating a management plan that defines how the project scope will be:

- Defined
- Validated
- Monitored
- Controlled



Plan Scope Management (Contd.)

The main purpose of plan scope management is to ensure that:

- Stakeholder requirements are properly managed
- Scope boundaries are clearly identified
- Changes to scope are controlled systematically
- The project team follows a consistent approach to scope management



Plan Scope Management (Contd.)

Inputs	Tools and Techniques	Outputs
• Project charter • Project management plan • Enterprise environmental factors • Organizational process assets	• Expert judgment • Meetings • Data analysis	• **Scope management plan:** A document that describes how the project scope will be defined, validated, monitored, and controlled throughout the project lifecycle. • **Requirements management plan:** A document that explains how project and product requirements will be collected, analyzed, documented, prioritized, tracked, and managed during the project.



Plan Scope Management (Contd.)

Example: University Learning Management System (LMS)

During scope planning, the project team decides:

- How student requirements will be gathered
- How new feature requests will be approved
- How deliverables will be validated by university management
- How scope changes will be documented

Key Processes in Scope Management



Collect Requirements

- **Collect Requirements** is the process of determining, documenting, and managing the needs and expectations of stakeholders to meet project objectives.
- This process helps the project team clearly understand what stakeholders expect from the product or service being developed and forms the foundation for defining project scope.



Collect Requirements(Contd.)

The main purpose of collect requirements is to:

- Understand stakeholder needs
- Define project and product requirements
- Establish a basis for project scope
- Reduce misunderstandings and rework
- Improve stakeholder satisfaction



Collect Requirements(Contd.)

Common Requirements Types:

- **Functional Requirements**
  - Describe what the system should do.
  - Example: User login functionality
- **Non-Functional Requirements**
  - Describe system qualities.
  - Example: System should respond within 2 seconds
- **Business Requirements**
  - Describe organizational goals.
  - Example: Improve online customer service
- **Technical Requirements**
  - Describe technical constraints.
  - Example: System must support cloud deployment



Collect Requirements(Contd.)

Inputs	Tools and Techniques	Outputs
• Project charter • Stakeholder register • Project management plan	• Interviews • Brainstorming • Workshops • Questionnaires and surveys • Observation • Prototyping • Focus groups	• **Requirements documentation:** A formal document that records stakeholder needs, expectations, and detailed project and product requirements collected during the project. • **Requirements traceability matrix (RTM):** A table used to track requirements throughout the project lifecycle to ensure each requirement is developed, tested, and delivered successfully.



Collect Requirements (Contd.)

Example: Online Food Delivery System

- Stakeholders:
  - Customers
  - Restaurant owners
  - Delivery riders
  - Administrators
- Collected requirements may include:
  - Online order placement
  - Real-time order tracking
  - Digital payment integration
  - Delivery notifications

Key Processes in Scope Management



Define Scope

- **Define Scope** is the process of developing a detailed description of the project and product by analyzing the collected requirements and stakeholder expectations.
- This process establishes clear project boundaries by identifying what is included and excluded from the project.



Define Scope (Contd.)

The main purpose of define scope is to:

- Create a detailed project scope statement
- Clarify project deliverables
- Define project boundaries
- Establish acceptance criteria
- Reduce misunderstandings and scope creep

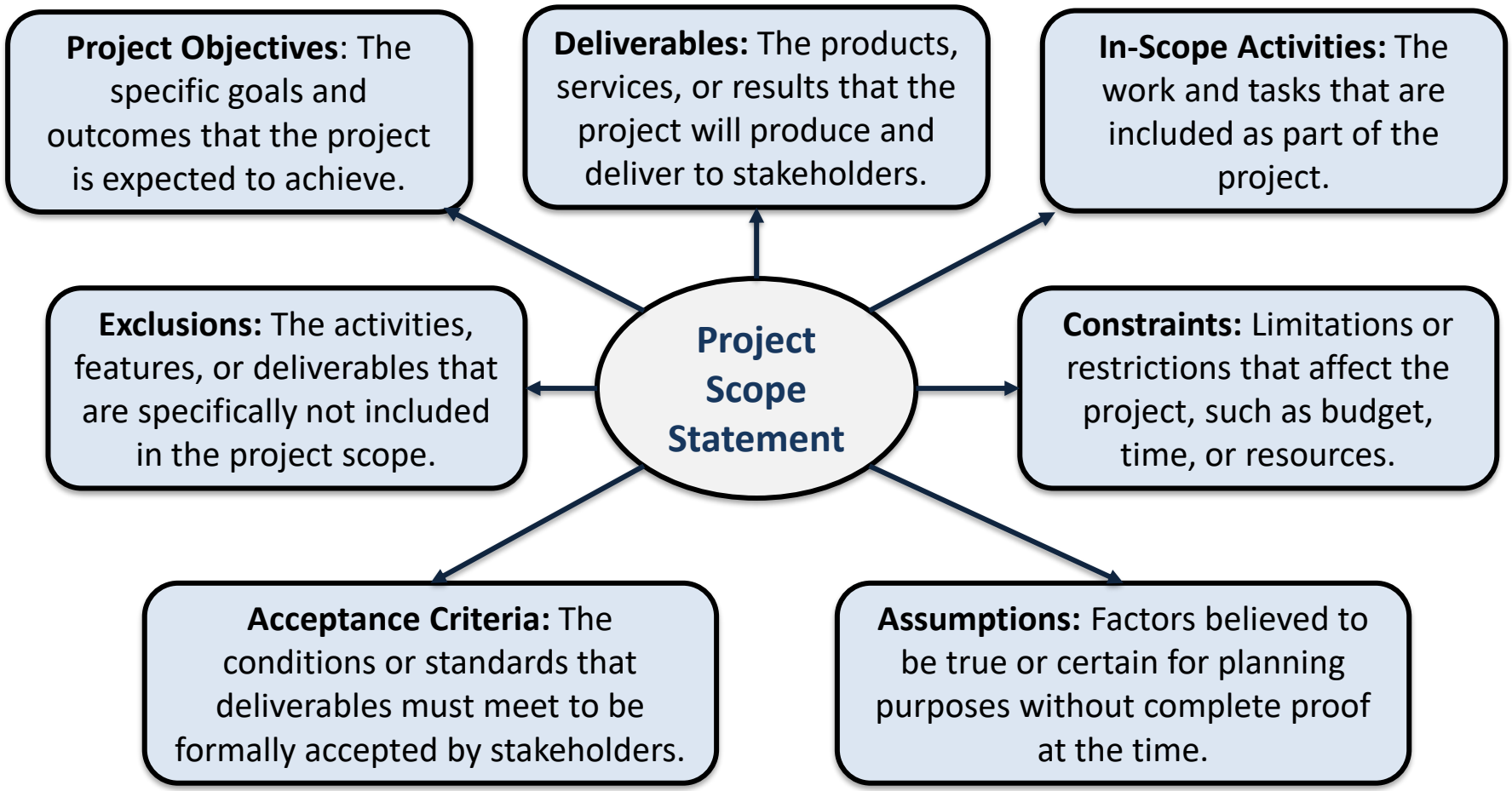


Define Scope(Contd.)

Inputs	Tools and Techniques	Outputs
• Project charter • Requirements documentation • Requirements traceability matrix • Organizational process assets	• Expert judgment • Product analysis • Alternatives analysis • Facilitation workshops	• **Project Scope Statement**: A detailed document that describes the project deliverables, boundaries, assumptions, constraints, and acceptance criteria for the project. **Project Document Updates**: Revisions made to existing project documents to reflect changes or additional information identified during the scope definition process.

Define Scope (Contd.)

Components of a Project Scope Statement





Define Scope (Contd.)

Example: Mobile Banking Application

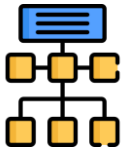
Included Scope

- User registration
- Balance inquiry
- Fund transfers
- Bill payments

Excluded Scope

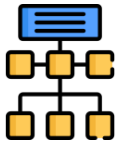
- Cryptocurrency transactions
- International money transfers

Key Processes in Scope Management



Create Work Breakdown Structure (WBS)

- **Create Work Breakdown Structure (WBS)** is the process of subdividing project deliverables and project work into smaller, more manageable components called work packages.
- The WBS helps organize and define the total scope of the project by breaking complex work into structured sections that are easier to plan, estimate, assign, monitor, and control.



Create Work Breakdown Structure

The main purpose of creating a Work Breakdown Structure is to:

- Organize project work systematically
- Improve cost and time estimation
- Clarify responsibilities
- Facilitate monitoring and control
- Ensure all required work is included

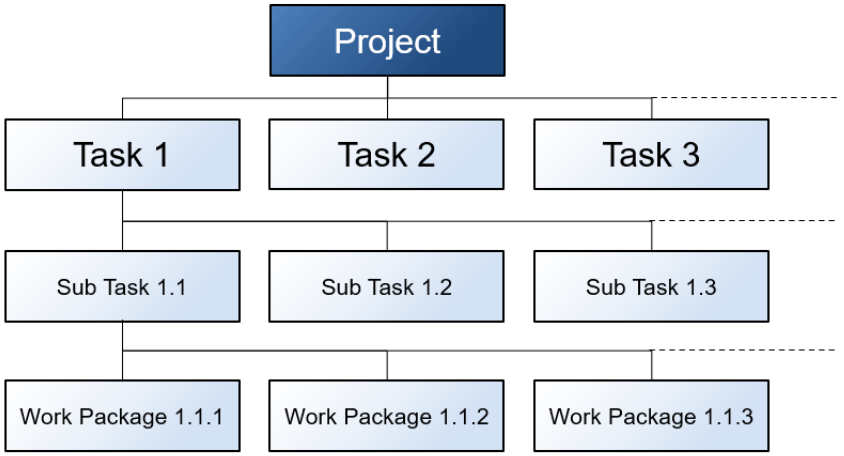
Create Work Breakdown Structure (Contd.)

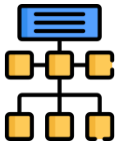
Key Concept – Decomposition

- The primary technique used in creating a WBS is **decomposition**, which means dividing large deliverables into smaller manageable work packages.
- Example:

  Website Development

  - Frontend Design
  - Backend Development
  - Database Setup
  - Testing
  - Deployment



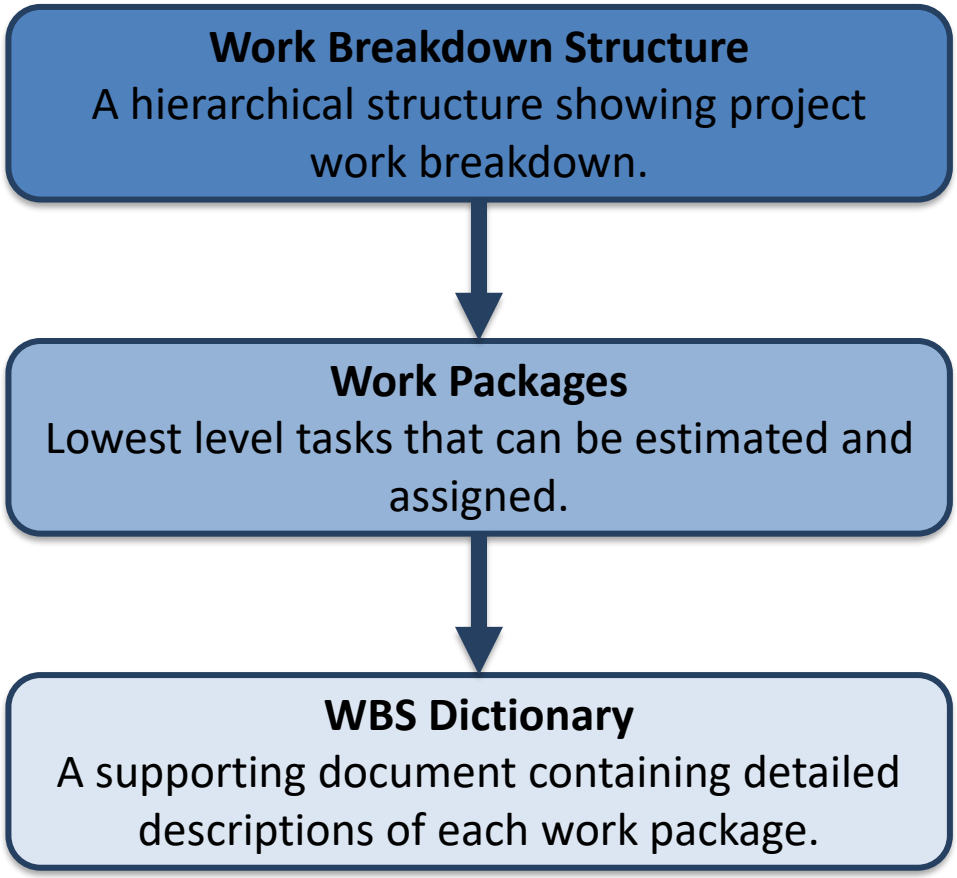


Create Work Breakdown Structure (Contd.)

Inputs	Tools and Techniques	Outputs
• Project scope statement • Requirements documentation • Organizational process assets	• Decomposition • Expert judgment	• **Work Breakdown Structure (WBS)**: A hierarchical decomposition of the total project work into smaller manageable components and work packages. • **WBS Dictionary**: A document that provides detailed information about each work package in the WBS, including descriptions, responsibilities, and estimates. • **Scope Baseline**: The approved version of the project scope statement, WBS, and WBS dictionary used as a basis for measuring project performance.

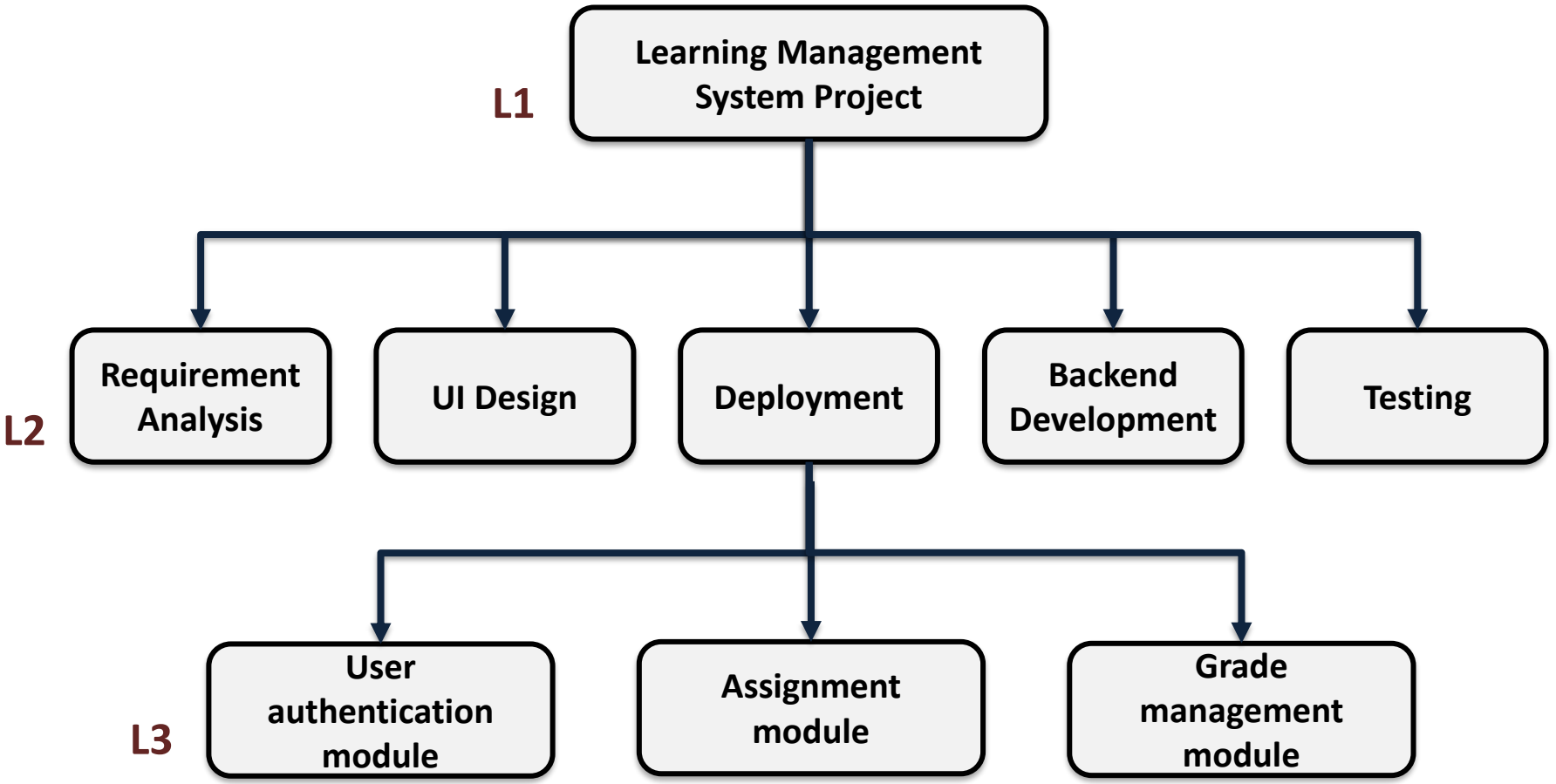
Create Work Breakdown Structure (Contd.)

Components of a Work Breakdown Structure



Create Work Breakdown Structure (Contd.)

Example: Online Learning Management System (LMS)



Key Processes in Scope Management



Validate Scope

- **Validate Scope** is the process of formally reviewing and accepting completed project deliverables by stakeholders or customers.
- This process ensures that the deliverables meet the agreed requirements and acceptance criteria defined during project planning.
- **Key Activities in Validate Scope**
  - Reviewing completed deliverables
  - Comparing results against acceptance criteria
  - Conducting inspections and demonstrations
  - Obtaining formal approval or feedback



Validate Scope (Contd.)

The main purpose of validate scope is to:

- Obtain formal stakeholder acceptance
- Confirm deliverables meet requirements
- Ensure project work aligns with scope
- Reduce disputes and misunderstandings

Validate Scope focuses on customer and stakeholder approval of completed work before moving to the next phase or closing the project.



Validate Scope (Contd.)

Inputs	Tools and Techniques	Outputs
• Verified deliverables • Scope baseline • Requirements documentation • Work performance data	• Inspection • Decision-making techniques • Meetings	• **Accepted Deliverables**: Completed project outputs that have been formally reviewed and approved by the customer or stakeholders. • **Change Requests**: Formal proposals for modifications to project scope, deliverables, schedule, or other project components identified during scope validation. • **Work Performance Information**: Processed and analyzed project performance data used to assess progress and support decision-making. • **Project Document Updates**: Revisions made to project documents to reflect changes, approvals, or new information identified during scope validation.

Validate Scope Vs Quality Control

Control Quality focuses on verifying that deliverables are technically correct and free from defects, while **Validate Scope** focuses on obtaining formal stakeholder or customer acceptance of those deliverables.

Validate Scope	Control Quality
Focuses on stakeholder acceptance	Focuses on correctness and quality
Done mainly by customers/stakeholders	Done mainly by quality team
Checks if requirements are met	Checks if deliverables are defect-free



Validate Scope(Contd.)

Example: University LMS Project

- Completed deliverables such as:
- Student login module
- Online assignment submission
- Grade viewing system
- are demonstrated to university management and lecturers for formal acceptance before final deployment.
- If stakeholders identify missing features, a change request may be raised.

Key Processes in Scope Management



Control Scope

- **Control Scope** is the process of monitoring the status of the project and product scope and managing changes to the scope baseline.
- This process ensures that all requested changes are properly evaluated, approved, and controlled so that the project remains aligned with its objectives, budget, and schedule.
- **Key Activities in Control Scope**
  - Comparing actual work against the scope baseline
  - Identifying scope variances
  - Reviewing change requests
  - Managing approved scope changes
  - Updating project documents



Control Scope (Contd.)

The main purpose of control scope is to:

- Monitor project scope performance
- Prevent uncontrolled scope changes
- Manage approved change requests
- Maintain alignment with project objectives
- Reduce scope creep

Control Scope helps ensure that only authorized work is performed during the project.



Control Scope (Contd.)

Inputs	Tools and Techniques	Outputs
• Project management plan • Requirements documentation • Requirements traceability matrix • Work performance data	• Data analysis • Variance analysis • Change control systems	• **Work Performance Information**: Analyzed project data that shows the current status of scope performance and helps support project decisions. • **Change Requests**: Formal requests proposing modifications to the approved project scope, deliverables, or scope baseline. • **Project Management Plan Updates**: Approved revisions made to components of the project management plan to reflect scope changes or corrective actions. • **Project Document Updates**: Modifications made to project documents to keep project information accurate and up to date after scope control activities.

Control Scope (Contd.)

Scope Creep

A major reason for Control Scope is to prevent **scope creep**, which refers to uncontrolled expansion of project scope without corresponding adjustments to:

- Cost
- Time
- Resources

Common Causes

- Poor requirement gathering
- Informal stakeholder requests
- Weak change control processes





Control Scope(Contd.)

Example: Mobile Banking Application

- During development, the client requests an additional feature for cryptocurrency transactions that was not included in the approved scope.
- The project team:
  1. Evaluates the impact on cost and schedule
  2. Submits a formal change request
  3. Obtains approval before implementation
- This process prevents uncontrolled feature additions and budget overruns.



Scope Management in Agile

- Agile scope management focuses on delivering value incrementally through continuous collaboration, adaptive planning, and flexible handling of changing requirements.
- **Key Characteristics of Agile Scope Management**
  - **Incremental Delivery:** Project work is delivered in small increments or iterations rather than as one final delivery.
  - **Flexible Requirements:** Requirements can change during the project lifecycle to meet evolving customer needs.
  - **Continuous Stakeholder Collaboration:** Customers and stakeholders regularly provide feedback to refine project scope.
  - **Prioritized Backlog:** Features and requirements are maintained in a product backlog and prioritized based on business value.

Agile Scope Management Components

- **Product Backlog**

  A prioritized list of features, enhancements, and requirements.

- **User Stories**

  Short descriptions of system functionality from the user's perspective.

  Example:

  "As a customer, I want to track my delivery order in real time."

- **Sprint Planning**

  Selection of backlog items to be completed during a sprint or iteration.

- **Iterative Reviews**

  Stakeholders review completed work frequently and suggest adjustments.



In Class Activity 1

Multiple Choice Question

Which statement BEST describes the purpose of a Work Breakdown Structure (WBS)?

A. To divide project work into manageable components

B. To estimate project risks

C. To calculate project budget variance

D. To approve completed deliverables



Multiple Choice Question

A client requests several additional features after the project scope has already been approved. What should the project manager do FIRST practice?

A. Immediately start development

B. Reject all new requirements

C. Evaluate the impact through scope control procedures

D. Increase the project budget automatically



What is Cost Management?

- **Project Cost Management** refers to the processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling project costs so that the project can be completed within the approved budget.



- Cost management ensures that project resources are used efficiently while delivering the expected business value and project objectives.

Main Objectives of Cost Management

- Estimate project expenses accurately
- Develop a realistic project budget
- Monitor project spending
- Control cost overruns
- Support effective financial decision-making

Effective cost management helps organizations complete projects successfully without exceeding financial limits.



Key Processes in Cost Management

- Following are the major cost management activities:


Plan Cost Management


Estimate Costs


Determine Budget


Control Costs

Key Processes in Cost Management



Plan Cost Management

- **Plan Cost Management** is the process of defining how project costs will be estimated, budgeted, managed, monitored, and controlled throughout the project lifecycle.

- This process establishes the policies, procedures, and documentation required for effective cost management and provides guidance to the project team on handling financial aspects of the project.



Plan Cost Management (Contd.)

The main purpose of plan cost management is to ensure that:

- Define the approach for managing project costs
- Establish cost estimation methods
- Set budgeting and reporting procedures
- Define cost control thresholds
- Ensure consistent financial management throughout the project



Plan Cost Management (Contd.)

Inputs	Tools and Techniques	Outputs
• Project charter • Project management plan • Enterprise environmental factors • Organizational process assets	• Expert judgment Data analysis Meetings	• **Cost Management Plan**: A document that describes how project costs will be planned, estimated, budgeted, monitored, and controlled throughout the project lifecycle.



Plan Cost Management (Contd.)

Key Components of a Cost Management Plan

- A Cost Management Plan may include:
  - Units of measurement
  - Level of accuracy and precision
  - Organizational procedures
  - Cost control thresholds
  - Reporting formats
  - Rules for performance measurement





Plan Cost Management (Contd.)

Example: Mobile Application Development Project

- The Cost Management Plan may define:
  - Developer hourly rates
  - Budget approval procedures
  - Monthly cost reporting methods
  - Cost variance thresholds
  - Software license budgeting approach

This helps the project manager monitor and control expenses effectively during development.

Key Processes in Cost Management



Estimate Cost

- **Estimate Costs** is the process of developing an approximation of the monetary resources required to complete project work and activities.

- This process helps the project team predict the expected costs of resources such as labor, materials, software, equipment, and services needed to successfully complete the project.



Estimate Cost (Contd.)

The main purpose of estimate cost is to ensure that:

- Predict project expenses
- Support budgeting activities
- Assist decision-making
- Provide a basis for cost control
- Improve financial planning accuracy

Accurate cost estimation helps organizations avoid budget overruns and resource shortages.



Estimate Cost (Contd.)

Common Cost Estimation Techniques

- **Analogous Estimating:** Uses historical data from similar projects.
- **Parametric Estimating:** Uses statistical relationships and formulas.
- **Bottom-Up Estimating:** Estimates individual tasks and aggregates them.
- **Three-Point Estimating:** Uses optimistic, pessimistic, and most likely estimates to improve accuracy.
  - Expected estimate formula:

$$E = \frac{6O + 4M + P}{6}$$

Where: O = Optimistic estimate, M = Most likely estimate,
P = Pessimistic estimate



Estimate Cost(Contd.)

Inputs	Tools and Techniques	Outputs
• Project management plan • Scope baseline • Project schedule • Risk register	• Expert judgment • Cost aggregation • Data analysis • Estimation software	• **Cost Estimates**: Quantitative assessments of the expected monetary resources required to complete project activities and deliverables. • **Basis of Estimates**: A document that explains how cost estimates were developed, including assumptions, methods, constraints, and levels of accuracy. • **Project Document Updates**: Revisions made to existing project documents to reflect new cost-related information identified during cost estimation.



Estimate Cost (Contd.)

Example: E-Commerce Website Development

Estimated costs may include:

- UI/UX design
- Backend development
- Cloud hosting
- Testing
- Security integration

Activity	Estimated Cost
UI Design	LKR 150,000
Backend Development	LKR 400,000
Testing	LKR 120,000

Total estimated project cost = LKR 670,000



In Class Activity 2

Multiple Choice Question

A project manager wants to establish rules for cost variance thresholds, reporting formats, and estimation accuracy before estimating project expenses. Which process is MOST directly responsible for this activity?

A. Determine Budget

B. Estimate Costs

C. Plan Cost Management

D. Control Costs



Multiple Choice Question

A software company estimates the development cost of a mobile application by separately estimating UI design, backend APIs, testing, deployment, and security integration before aggregating them into a total estimate. Which estimation technique is MOST likely being used?

A. Analogous Estimating

B. Parametric Estimating

C. Three-Point Estimating

D. Bottom-Up Estimating



Short Answer Question

Why might a project team use Three-Point Estimating instead of relying on a single cost estimate in software projects?



Key Processes in Cost Management



Determine Budget

- **Determine Budget** is the process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline for the project.

- This process combines all approved cost estimates, contingency reserves, and funding requirements to create the total project budget used for measuring and controlling project performance.



Determine Budget (Contd.)

The main purpose of determine budget is to ensure that:

- Establish the approved project budget
- Create a cost baseline
- Allocate funding requirements
- Support project financing and cost control
- Provide a basis for measuring project performance

A well-defined budget helps ensure that the project has sufficient financial resources to complete planned work successfully.



Determine Budget(Contd.)

Key Components of the Project Budget

- **Activity Cost Estimates:** Estimated costs for project tasks and work packages.
- **Contingency Reserves:** Funds allocated for identified project risks and uncertainties.
- **Management Reserves:** Additional funds reserved for unforeseen work outside the project baseline.
- **Cost Baseline:** Approved version of the time-phased project budget used for performance measurement.

Cost Baseline = Estimated Project Costs + Contingency Reserves

Note: Management reserves are NOT included in the cost baseline.



Determine Budget (Contd.)

Inputs	Tools and Techniques	Outputs
• Cost estimates • Basis of estimates • Project schedule • Risk register • Resource calendars	• Cost aggregation • Historical information review • Funding limit reconciliation • Expert judgment	• **Cost Baseline**: The approved version of the project budget, excluding management reserves, used to measure and monitor project cost performance. • **Project Funding Requirements**: The total funding needed for the project, including planned expenditures and reserve allocations over time. • **Project Document Updates**: Revisions made to project documents to reflect budget-related changes and updated financial information identified during budgeting activities.



Determine Budget (Contd.)

Example: Online Banking System

Category	Cost
Software Development	LKR 2,000,000
Testing	LKR 500,000
Cloud Hosting	LKR 300,000
Training	LKR 200,000
Contingency Reserve	LKR 250,000

Total Project Budget: LKR 3,250,000

This approved budget becomes the baseline for monitoring project spending during execution.

Key Processes in Cost Management



Control Costs

- **Control Costs** is the process of monitoring project spending, managing changes to the cost baseline, and ensuring that project costs remain within the approved budget.
- **Key Activities in Control Costs:**
  - Tracking actual project spending
  - Measuring cost performance
  - Performing variance analysis
  - Managing change requests
  - Updating budget forecasts



Control Costs (Contd.)

The main purpose of control costs is to ensure that:

- Monitor project expenditures
- Compare actual costs with the cost baseline
- Identify cost variances
- Prevent budget overruns
- Manage approved budget changes
- Forecast future project costs

Effective cost control helps ensure the project delivers value while staying within financial limits.



Control Costs (Contd.)

Earned Value Management (EVM)

- A common technique used in Control Costs is **Earned Value Management (EVM)**.
- **Key Metrics**
  - **Planned Value (PV):** Authorized budget for scheduled work.
  - **Earned Value (EV):** Value of completed work.
  - **Actual Cost (AC):** Actual amount spent on completed work.
- **Cost Variance Formula**

$$CV = EV - AC$$

CV > 0 → Under budget **CV < 0 → Over budget**

- **Cost Performance Index**

$$CPI = \frac{EV}{AC}$$

CPI > 1 → Good cost performance **CPI < 1 → Poor cost performance**



Control Costs (Contd.)

Inputs	Tools and Techniques	Outputs
• Project management plan • Project funding requirements • Work performance data • Organizational process assets	• Earned value analysis • Forecasting • Variance analysis • Trend analysis • Project management information systems	• **Work Performance Information**: Analyzed cost and schedule data that indicates the current status and performance of the project budget. • **Cost Forecasts**: Predictions of future project costs and expected final budget outcomes based on current project performance. • **Change Requests**: Formal proposals for modifications to the project budget, cost baseline, or related project components. • **Project Management Plan Updates**: Approved revisions made to the project management plan to reflect cost control decisions and changes. • **Project Document Updates**: Modifications made to project documents to keep cost-related information accurate and up to date.



Control Costs (Contd.)

Example: Mobile Application Development

- Planned development cost = LKR 1,000,000
- Actual spending after 2 months = LKR 1,250,000
- The project manager identifies:
  - Higher-than-expected cloud service costs
  - Additional testing expenses
- Corrective actions may include:
  - Reducing non-essential features
  - Optimizing cloud usage
  - Revising resource allocation



Cost Management in Agile

- Agile cost management focuses on continuously monitoring and controlling project costs while delivering value incrementally through iterative development and adaptive planning.
- **Key Characteristics of Agile Cost Management**
  - **Iterative Budgeting:** Costs are managed sprint-by-sprint or iteration-by-iteration rather than through a single fixed plan.
  - **Value-Based Spending:** Project spending is prioritized based on delivering the highest business value first.
  - **Flexible Scope with Controlled Cost:** In Agile projects, time and team size are often fixed, while scope can change depending on priorities and feedback.
  - **Continuous Monitoring:** Costs are reviewed regularly during sprint reviews, retrospectives, and progress meetings.

Common Agile Cost Management Approaches

- **Team-Based Budgeting**

  Budgeting is often based on stable Agile teams working over a fixed time period.

  Example:

  - Scrum team monthly cost = LKR 1,500,000
  - Project duration = 6 months
  - Estimated project cost = LKR 9,000,000

- **Velocity-Based Forecasting**

  Future project costs are estimated using team velocity and backlog completion rates.

- **Burn Rate Tracking**

  Monitoring how quickly project funds are being spent over time.



In Class Activity 3

Multiple Choice Question

A project manager aggregates activity cost estimates and contingency reserves to create a performance measurement baseline for the project. Which statement is MOST accurate regarding this baseline?

A. It includes management reserves for unknown risks

B. It is mainly created during Control Costs

C. It is used to measure project cost performance during execution

D. It represents only direct labor costs of the project



Multiple Choice Question

A software project has the following values after four months:

- **Earned Value (EV) = LKR 4,500,000**
- **Actual Cost (AC) = LKR 5,000,000**

Which interpretation is MOST accurate?

A. The project is under budget because actual work completed is high

B. The project is over budget because the actual spending exceeds earned value

C. The project performance cannot be determined without Planned Value

D. The project has a positive cost variance



Short Answer Question

Why is cost forecasting important during the Control Costs process in software projects?



Summary

- Scope Management ensures the project includes only the required work.
- Main scope activities include planning, defining, validating, and controlling scope.
- WBS breaks project work into smaller manageable tasks.
- Scope control helps prevent scope creep and unmanaged changes.
- Cost Management focuses on estimating, budgeting, and controlling project expenses.
- Cost estimation techniques include Analogous, Bottom-Up, and Three-Point Estimating.
- Cost Baseline and Earned Value Management (EVM) are used to monitor project cost performance.
- Agile projects manage scope and cost iteratively through sprints, backlogs, and continuous stakeholder feedback.